Week 6 — Demo: Magnetic Susceptibility of Rocks

Geophysics 3A: Potential Fields & Geothermics

May 27, 2026

Purpose

This activity links magnetic susceptibility to mineralogy and geological processes such as alteration, metamorphism, and weathering.

Key Reminder

Magnetic susceptibility (χ) describes how strongly a material becomes magnetised:

$$\mathbf{M} = \chi \mathbf{H}$$

Samples and Observations

Paired Comparisons

Peridotite vs Serpentinite

How does serpentinisation lead to high magnetic susceptibility?

Fresh Gabbro vs Altered Basalt

Why can alteration dramatically reduce susceptibility in mafic rocks?

Sample	Rock type	Geological context	Susceptibility (SI)	Notes
1				
2				
3				
4				
5				
6				
7				
8				

Granite or Sedimentary Protolith vs Metamorphosed Equivalent

Which minerals control susceptibility in felsic or sedimentary rocks?

Synthesis

1. Why is susceptibility not a simple proxy for bulk composition?
2. Which geological processes increase susceptibility?
3. Which decrease it?

Reflection

Explain why petrology is essential for interpreting magnetic anomalies.